Student Number : 24 334 799

UNIVERSITÉ
DE GENÈVE

Governing & Designing Organizations for Stakeholders - Iraj Nayebkhil

Structuring for Growth:
How Organisational Design Decisions Shape Start-up Performance and Growth Trajectories

I. Introduction

Start-ups have become a cornerstone of innovation in the 21st century. With their strategic pursuit of scalable business models and the immense disruptive potential that they hold, start-ups play a pivotal role in job creation, technological advancement, and reshaping industries. Start-ups are also particularly interesting as their goals and, consequently, strategies differ from traditional small and medium-sized enterprises (SMEs) or micro-enterprises (MSEs), which often prioritize stability and incremental growth as start-ups typically aim for rapid expansion, high-risk ventures, and global scalability. This distinction also makes the internal decisions they make, **especially about organisational structure, a key factor in their early and subsequent performance**.

For this paper, the core question guiding this study is: ***How do organisational structure decisions impact the performance and growth trajectories of start-ups?*** To answer this, a structured literature review will be conducted, synthesizing existing research across disciplines such as entrepreneurship, organisational theory, and strategic management. The review will explore the relationship between structural design and key performance outcomes like adaptability, speed of execution, and long-term scalability.

II. Methodology

This study employs a **narrative literature review** approach to explore how organizational structure decisions influence the performance and growth trajectories of start-ups. To ensure academic rigor and relevance, the literature selected for review meets the following criteria:

- **Peer-reviewed journal articles** were prioritized to maintain academic credibility.
- Studies had to **explicitly address start-ups** as their unit of analysis or discuss organizational structure in contexts relevant to start-up environments.
- Only papers that explored **organizational structure, internal design choices, and their impact on growth or performance** were included.
- Articles focused on unrelated themes such as business model design, product innovation, or market performance were **excluded** unless they provided direct structural insights.

Once selected, the papers were reviewed and grouped by themes such as types of organizational structure, structural evolution, impacts on agility and coordination, and correlations with performance outcomes like time-to-market or funding success. The findings were then synthesized to identify recurring patterns, theoretical frameworks, and knowledge gaps, providing the foundation for the analysis. As for the timeframe, the review primarily focuses on literature published in the **last 10 to 15 years (2010–2025)**, covering the modern digital and start-up context. The preceding discussion will interpret these findings through the lens of established organizational design theories and will also highlight gaps in current research.

III. Literature Review

Let us first start off by summarizing the key organizational structures commonly discussed in the context of start-ups. At the most basic level, **flat and hierarchical structures** represent two ends of the spectrum. Flat structures are typical of early-stage start-ups, where teams are small, roles are fluid, and decision-making is fast and often centralized around the founders. In contrast, hierarchical structures introduce multiple layers of authority, clearer reporting lines, and greater formalization, features that tend to emerge as start-ups grow and require more coordination and accountability.

Beyond the flat–hierarchical distinction, start-ups may adopt more specialized organizational forms as they grow. Four common structures are **functional**, **divisional**, **matrix**, and **holacracy**. A **functional structure** organizes the company based on specialized roles or departments such as marketing, product development, finance, and operations. It is one of the most common early formalizations for start-ups **transitioning from flat models**, especially in tech. A **divisional structure** groups teams according to products, customer segments, or geographic markets. This model enables **greater focus, accountability, and responsiveness within each division**, making it particularly useful for start-ups expanding into new markets or offering multiple product lines. The **matrix structure** combines elements of both functional and divisional models. Employees typically report to two managers: one by function and another by project or product. This design encourages **collaboration and knowledge sharing** across departments while maintaining specialized roles Finally, **holacracy** represents a non-traditional, decentralized alternative. Popularized by companies like Zappos, holacracy eliminates formal hierarchies and replaces them with **self-organizing**

teams ("circles") that hold distributed authority. Each team is responsible for its own roles and decisions.

Now that we have a good idea of the major organizational structures and how they are typically defined, let’s see what some studies have said about how these structures play out in real start-up environments. One particularly influential empirical study on the evolution from informal to formal structures is by **Grimpe, Murmann and Sofka (2019)**. Based on a multi-year analysis of high-technology start-ups, their research showed that as firms move from the start-up stage into stages of growth and expansion, informal structures, such as overlapping roles and ad hoc decision-making, begin to hinder coordination and scalability. The study found that introducing formalized structures (clear reporting lines, departmentalization, and role specialization) was positively associated with improved operational efficiency and sustained innovation. Interestingly, Grimpe, Murmann and Sofka observed that these transitions often followed identifiable “structural turning points,” typically triggered by increased employee headcount, product diversification, or new rounds of external funding. This suggests that organizational structure is not static but evolves in response to internal complexity and strategic milestones.

Furthermore, empirical evidence from a study by **Wulf (2012)** highlights how flatter organizations tend to outperform their hierarchical counterparts in environments that require flexibility and innovation. Their research, based on over 4,000 firms across multiple countries, found that **firms with fewer layers of management and greater decentralization were significantly more likely to introduce new products and adopt process innovations**. However, the study also noted that as firms scale, some degree of hierarchy becomes beneficial for managing complexity and maintaining operational clarity, suggesting that the optimal structure may shift over time depending on the growth stage and strategic focus of the company. It is important to acknowledge that **flatness still holds value**, particularly in environments where innovation speed and rapid iteration are paramount. A complementary study by **Xu, Wu, and Evans (2022)** on scientific teams found that flatter structures often yield **more novel and disruptive work**, particularly in knowledge-intensive or research-driven settings. This suggests that **the effectiveness of structure is highly context-dependent**, what works in a lab or early-stage tech start-up may not translate well to a scaling, market-facing company.

Building on these insights, several empirical studies have examined how different organizational structures influence core performance drivers such as **speed, innovation, and coordination**. For instance, a study by **Lee (2022)** examined the impact of organizational structure on start-up performance. The research found that flatter structures, characterized by fewer hierarchical layers, facilitated faster decision-making. This was attributed to the freer flow of information and greater autonomy afforded to front-line teams. Such agility proved particularly critical for start-ups operating under time pressure or in rapidly changing conditions. In terms of innovation, a more recent empirical study by **Keum and See (2017)** offers updated insights into the role of structure. Their research found that **organic structures, characterized by decentralization, flexibility, and low formalization, were positively associated with higher levels of exploratory and adaptive innovation**, particularly in dynamic and uncertain environments. These structures allowed for broader idea generation, open communication, and autonomy, which are crucial for early-stage innovation. Conversely, **mechanistic structures**, marked by rigid hierarchies and formalized processes, were found to support **coordination and implementation efficiency**, especially during later stages of scaling or in industries where precision and predictability are critical. This aligns with earlier findings and reinforces the notion that the structural context must match the innovation objectives and stage of the start-up. A supporting study by **Piezunka and Dahlander (2015)**, focusing on European start-ups, found that those with moderate levels of formal structure achieved higher operational efficiency, interestingly enough, *without* sacrificing innovation, highlighting the importance of **balance rather than extremes**.

While structural evolution is essential for scaling, it often introduces significant challenges that can undermine a start-up's cohesion and performance. The study by **Grimpe, Murmann and Sofka (2019)** also found that the introduction of middle management and formal processes in high-tech start-ups can lead to a **loss of the entrepreneurial spirit and agility** that characterized the early stages of the company. This shift often results in decreased employee motivation and engagement, as the informal communication and decision-making processes are replaced by more rigid structures. Similarly, research by **Wasserman (2012)** in his book indicates that as start-ups grow and attract external investors, founders often face pressure to relinquish control to professional managers. This transition can be challenging, as founders may struggle to adapt to new roles or resist changes that they perceive as threats to their vision. In parallel to internal role evolution, the composition of the board of directors also changes over time. Early-stage boards are typically composed of founders and initial investors, but as start-ups mature, they often bring in independent directors to enhance governance. A large-scale analysis of over 7,200 startups found that board independence improved conflict resolution between founders and investors and positively impacted decision quality but the resulting power struggles might hinder decision-making and negatively impact company performance (Kroll, Walters and Wright, 2008). Thus, for many start-ups, this creates a structural paradox: **the need to introduce discipline and coordination to support scaling,** while also preserving the entrepreneurial drive and culture that fueled early success. These studies demonstrate that structure alone is not a 'remedy'; it must be implemented thoughtfully, considering timing, team dynamics and cultural continuity.

In addition to this, the study by **Liu et al. (2021)** also talks about the need to be flexible by discussing the trade-offs between centralization and decentralization of organizational structures. While decentralization fosters innovation and adaptability, it can also lead to coordination challenges. Conversely, centralization can enhance efficiency and consistency but may stifle creativity and responsiveness. Start-ups however, use these structural differences to their advantage as an adaptation mechanism. Research by **Butticè, Colombo and Rovelli (2024)** analysing 241 Italian startups reflects the recurring argument that **centralization is often effective in early stages**, where founders must make rapid, cohesive decisions and maintain tight control over strategy and culture. In this phase, centralized decision-making reduces coordination costs and ensures that the limited team aligns around a shared vision. However, as start-ups grow in size and complexity, **decentralization becomes increasingly valuable**. A study by **Cattani, Ferriani, and Lanza (2017)** found that start-ups in knowledge-intensive industries, such as tech or biotech, benefited from decentralization because it allowed specialized teams to make decisions closer to the problem, enhancing innovation responsiveness and time-to-market. For instance, Tesla's evolution is a good example that illustrates this structural transition in a real-world context. Initially operating with a flat hierarchy that enabled rapid innovation and close founder control, Tesla later adopted a functional structure with clear departmentalization across engineering, finance, and sales. As the company scaled, it embraced a unique model, functioning as a "startup of startups" within a larger corporate framework. This approach allowed it to retain agility in emerging divisions like AI and energy storage, while benefiting from centralized coordination (Meyer, 2024). Tesla's trajectory exemplifies how structural adaptation is not linear but can be hybrid and dynamic, especially in innovation-driven industries. Nevertheless, **Gulati and Puranam (2010)** makes the holistic argument that a hybrid structure, centralizing functions like finance or legal for efficiency, while decentralizing product teams for agility, is often optimal during the scaling phase. This balance allows start-ups to retain innovative capacity while building the managerial depth needed for operational discipline. Therefore, the decision to centralize or decentralize should not be treated as binary, but rather as a **strategic continuum** that evolves with the start-up's size, industry demands, and growth stage.

While there is limited research on how specialized organizational forms beyond the flat–hierarchical spectrum directly affect the performance of start-ups, two notable studies, **a study by Jack and Bayo (2024)** on holacracy and **a study by Burns and Wholey (1993)** on matrix management, connect to our previous arguments by shedding light on how these evolving structures perform in practice (there were no recent studies). The study on **Holacracy** and Organizational Performance analyzes 15 case studies and finds that holacracy can enhance agility, employee engagement, and innovation through decentralized authority. However, in SMEs, its success is highly dependent on a strong culture of transparency and readiness for self-management. Where such alignment is lacking, holacracy led to confusion and dips in productivity due to overlapping roles and unclear accountability. Meanwhile, the study on **Matrix Structures** investigates matrix adoption in U.S. hospitals but offers insights applicable to SMEs. It found that matrix structures are often adopted due to external pressures, such as mimetic or normative forces, but frequently abandoned due to internal challenges like dual-reporting confusion and control conflicts. For SMEs, these findings suggest that **while holacracy and matrix models can offer increased collaboration and responsiveness, they demand a high level of structural maturity**. Premature implementation, especially during early or transitional growth stages, can overload small teams and reduce clarity, undermining performance. While these findings are insightful, it is important to emphasize that the empirical basis for assessing non-traditional structures like holacracy and matrix models in start-ups remains limited. Much of the evidence comes from studies conducted in SMEs or non-start-up sectors such as healthcare, which differ in terms of scale, innovation cycles, and resource constraints. Therefore, although these models offer theoretical advantages like agility and cross-functional collaboration, any generalization to start-up contexts must be made cautiously.

To tie these insights together, a study by **Furr and Shipilov (2019)** offers a unifying perspective that reframes the apparent trade-offs between structure, innovation, and growth. Investigating over 30 high-growth start-ups across Europe and the U.S., their research found that the most successful firms didn’t simply transition from flat to hierarchical models or from informal to formal systems. Instead, they embraced what the authors term **“structured chaos”**, an intentional design where certain parts of the organization (such as HR, finance, and compliance) were tightly structured, while others (especially product development and R&D) remained fluid and autonomous. This **ambidextrous structuring** allowed start-ups to scale efficiently while maintaining the speed, creativity, and culture that drove their early success. Rather than viewing structure as a constraint, these firms used it as a strategic lever, **introducing just enough formalization to support growth without stifling innovation**. This study delivers a compelling insight: structure and agility are not mutually exclusive. When thoughtfully designed, organizational structure can act not as a limitation, but as an enabler of sustained performance and adaptability.

Now that we have explored how structural choices shape internal dynamics and innovative potential, we delve into how these decisions translate into concrete performance outcomes, such as funding acquisition, team productivity, and speed to market, which are often make-or-break factors for start-ups. Recent empirical research has made attempts to quantify these **direct impacts of organizational structure on key performance outcomes**. **Eisenmann (2020)** conducted a large-scale analysis of 470 early-stage start-ups and found a **strong positive correlation between early organizational structuring and funding success**. Start-ups that implemented basic formal systems, such as clear roles, accountability mechanisms, and early-stage operational routines, were more likely to attract venture capital. The study argues that visible structure signals professionalism and reduces perceived risk for investors, particularly in competitive funding environments. In terms of **team productivity**, a study by **Bai, Feng, and Yue (2017)** examined cross-functional teams in Chinese high-tech firms and found that more organic and functionally integrated organizational structures significantly improved internal alignment and overall team performance. The research shows that the presence of defined departmental roles reduced task redundancy and miscommunication—key barriers to productivity in fast-growing start-ups. Similarly, **a study by Colombo and Grilli (2010)** found that early adoption of formal organizational practices, such as clear role definitions and structured routines, was positively associated with growth in high-tech start-ups. Their findings suggest that such formalization can support better coordination and scalability. We can infer that these are critical for accelerating development and execution in competitive environments and thus, **reducing time-to-market.**

IV. Discussion

The reviewed literature strongly suggests that while flat structures are highly effective during a start-up's formative stages, they become less suitable as organizational demands evolve. This marks structure not as a fixed asset, but a dynamic lever that must shift in response to complexity. In the early stages, flat and informal structures are beneficial precisely because they minimize barriers to communication and accelerate decision-making. Start-ups often face uncertain markets, limited resources, and rapid iteration cycles, contexts where agility and founder-led coordination are critical to survival and innovation (Wulf, 2012; Lee, 2022). As teams expand, products diversify, and external accountability (like investors) increases, the absence of formalized structure can lead to inefficiencies, role confusion, and slower execution (Grimpe, Murmann and Sofka, 2019). Importantly, the solution is not to leap to rigid bureaucracy. Studies such as Keum and See (2017) and Piezunka and Dahlander (2015) show that a moderate degree of formalization, like introducing functional departments, can boost coordination and accountability **without compromising** innovation.

What is particularly notable across the reviewed literature is the consistent **preference for informal, flat, and less hierarchical structures** in supporting speed and innovation, two outcomes that are especially critical during the early stages of a start-up's development. These structures are credited with enabling rapid decision-making, fluid communication, and entrepreneurial attitude, which align directly with the strategic priorities of start-ups in their early-stages. However, it is important to recognize that flat structures, while often praised for enabling agility and innovation, are not without limitations. Wulf (2012) offers a critical view by demonstrating that the performance benefits of flatness are frequently overstated. In some cases, the absence of hierarchy leads to ambiguity in decision-making, bottlenecks around founders, and unclear accountability, particularly when start-ups begin to scale, in which case the literature tends to favor **more hierarchical structures**. This pattern is especially insightful: **the evolution from flat to more formalized forms is not just common, it reflects a deliberate shift in organizational priorities as start-ups transition from exploration to execution**. Thus, the key lies in evolving structure in proportion to organizational scale and needs. This reveals that **organizational structure decisions have a direct impact on start-up performance and growth** by influencing how effectively a start-up can adapt to its stage of development. Flatness enables exploration and speed; formalization enables coordination and scaling. Neither is superior in all cases, effectiveness depends on strategic alignment with the start-up's maturity and environment.

While the benefits of balancing flatness and formalization are well-supported, some start-ups experiment with even more complex and non-traditional structures in pursuit of sustained innovation and adaptability. These designs often go beyond the flat–hierarchical spectrum, aiming to embed flexibility and collaboration more deeply into the organization's DNA. For example, the study by Jack and Bayo (2024) on **holacracy** suggests that decentralized, authority-distributing models can enhance innovation, responsiveness, and employee engagement—**but only when supported by a culture of transparency and readiness for self-management**. In contrast, Burns and Wholey (1993) explore **matrix structures** and highlight their vulnerability to internal confusion and coordination breakdowns, especially when applied prematurely or without strong systems to manage dual reporting lines. These findings underscore that while advanced structural models can offer strategic advantages, they also introduce complexity that must be carefully timed and culturally supported to avoid undermining performance. Thus, we can infer that non-traditional structures like holacracy and matrix models offer potential benefits in adaptability and engagement, but their effectiveness depends heavily on internal readiness, particularly culture, communication, and leadership. These models are not easily transferable and, if adopted too early in a start-up's lifecycle, can create confusion and strain coordination, structural innovation must align with a start-up's maturity and evolve gradually to avoid undermining performance.

The above findings on advanced structural models highlight a recurring challenge: how can start-ups maintain the agility and creative spark of their early days while introducing enough structure to support growth?, precisely the tension that Furr and Shipilov (2019) capture through their concept of **"structured chaos."** Rather than fully formalizing or remaining entirely fluid, high-growth start-ups often adopt a **hybrid approach**, introducing structure in core functions like finance or HR while preserving flexibility in areas like R&D or product development. This model helps explain how start-ups can scale **without sacrificing innovation**. Furthermore, the importance of structure is not just conceptual; it influences concrete outcomes. Eisenmann (2020) demonstrates that early structural clarity improves **funding success**, while studies by Bai et al. (2023) and Colombo and Grilli (2010) show that defined roles and processes can enhance **productivity** and speed up **time-to-market**. Still, this evolution is not without risks. While many studies highlight the benefits of transitioning toward more formalized structures, there are notable cases where such shifts have led to negative consequences. For instance, Grimpe et al. (2019) found that middle management layers can dilute the entrepreneurial spirit, while Wasserman (2012) describes the internal friction that can arise when founders lose control to external managers, resulting in cultural misalignment, lowered employee morale, and even strategic drift. Therefore, structural change should not be viewed as inherently beneficial; its success hinges on the organization's cultural readiness, timing, and the way transitions are communicated and implemented. Taken together, the literature underscores that **organizational structure is neither fixed nor one-size-fits-all**, it must evolve with the start-up's goals to truly enhance performance.

With regards to the organizational structures impact, several important gaps remain in the literature. First, there is a notable lack of **post-COVID studies** that account for how remote work, digital collaboration tools, and shifting business environments may have influenced structural decisions in start-ups. Second, while flat and hierarchical structures are widely discussed, there is limited empirical research specifically examining **non-traditional forms like matrix structures or holacracy**, and even less on **functional and divisional models**. Most references to these complex models remain conceptual. Most importantly, there is an evident absence of rigorous, **empirical studies that track successful start-ups through multiple growth stages**, analyzing how and why they transitioned from one structural form to another. Existing insights are mostly drawn from isolated case studies or theory.

V. Conclusion

This paper set out to answer the question: *How do organisational structure decisions impact the performance and growth trajectories of start-ups?* The literature reviewed offers clear and consistent evidence that organizational structure decisions impact the performance and growth trajectories of start-ups in complex, context-dependent ways. While flat structures tend to support innovation, speed, and adaptability in the early stages, and more formalized structures promote coordination and scalability in later phases, the "right" structure is not universal. Its effectiveness depends heavily on the start-up's specific goals, which can vary significantly across industries and even between individual ventures. A biotech start-up might focus on regulatory approval may require formal structure earlier than a tech start-up aiming for viral growth. Similarly, a start-up that prioritizes creative autonomy and rapid experimentation may benefit from sustained decentralization, while one targeting operational efficiency and investor readiness may formalize more quickly. In short, structure is not a fixed formula but a strategic choice that must align with a start-up's evolving objectives, resource constraints, and market conditions. Start-ups that succeed are not those that adopt a particular structure, but those that **adapt structure intentionally** in line with their mission, stage, and environment.

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